Rank contraction: complete written costs

Human Compression project

Private numerical candidate; distinct review pending

The original is Brooksbank–Leemans, *Rank reduction of string C-group representations*, arXiv:1812.01055v2, 3 May 2019. The accepted scope contains 22 source clauses and 20 proof contexts. This comparison measures the complete source manuscript and the complete common paper and application, with no free overlap with earlier Topic B theory. It makes no new mathematical or source-credit claim.

Written papers

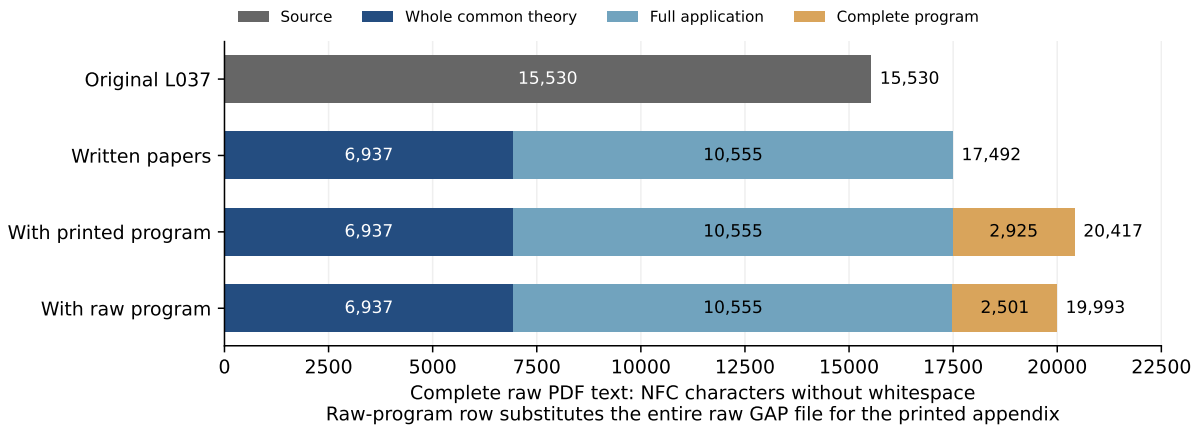
The primary unit is the number of non-whitespace characters in the complete raw PDF extraction, normalized to NFC. Titles, author blocks, running furniture, references and the application's references-only final page are included.

Complete document	Pages	Characters
Original L037	6	15,530
Whole common theory U	3	6,937
Application A	5	10,555
Written replacement $U + A$	8	17,492
Complete printed program	1	2,925

The combined written comparison is

$$\frac{c(L037)}{c(U) + c(A)} = \frac{15530}{17492} = 0.887834.$$

A value below one means that the replacement is longer in this coordinate. Here the complete written replacement has 1,962 more characters than the source.



Including the complete program

The GAP file `check_examples.g` is required for the displayed matrix and three literal A_{11} examples and includes doubled-edge validation. The uniform rank-contraction theorem and its infinite families do not depend on its output. The complete printed appendix costs 2,925 characters; the whole raw program costs 2,501 characters. These are alternatives:

$$\frac{c(L037)}{c(U) + c(A) + c(\text{printed program})} = \frac{15530}{20417} = 0.760641,$$

$$\frac{c(L037)}{c(U) + c(A) + c(\text{raw program})} = \frac{15530}{19993} = 0.776772.$$

Neither row combines the printed and raw versions. No line or diagnostic part is pruned from the whole file.

After acquiring the theory and program

Only after acquiring *all* of U and the complete declared program is the conditional application comparison

$$\frac{c(L037)}{c(A)} = \frac{15530}{10555} = 1.471341.$$

The prior acquisition costs are $6,937 + 2,925 = 9,862$ characters using the printed program, or $6,937 + 2,501 = 9,438$ using the raw alternative. This conditional ratio does not assert that the complete replacement is shorter.

Separate costs and limits

The complete operational driver has 6,950 raw bytes and 6,114 primary non-whitespace characters. The complete Lean file has 1,727 raw bytes and 1,280 primary characters. Its four declarations are scoped group-algebra facts; they do not formalize the full contraction theorem, its classical seeds or the source. These files are separately inventoried and are not silently included in the program rows. The mathematical GAP file has 2,680 raw bytes.

The end criterion and stated highest-rank classical constructions, orthogonal identifications and the prior A_{11} exhaustion remain retained inputs. Their full proof costs, historical software and full formalization costs are unknown, not zero. Conjecture 5.1 remains an open research target; no answer or question credit is claimed.

The source's three diagrams contain 264 XYCIRC glyph events. Every variant retains their literal extracted `/c16` through `/c23` strings. Those strings are not a complete measurement of the drawn graphs. Formula and graph layout costs also remain unknown; no full-visible-text or intrinsic mathematical-complexity claim is made.

The numerical author separately reviewed the accepted inputs but is the author of these new calculations, not their independent numerical reviewer. Publication requires a distinct review. No GAP or Lean computation was repeated for these numbers.

Metric and extraction sensitivity

All 30 coordinates use the same whole-document boundary. The five units are computed under NFC and NFKC. “Printed” and “Raw” include the corresponding whole program representation; “Conditional” assumes the full prior acquisition specified above. The JSON and CSV files retain all 120 exact fractions; the table rounds only the display.

Complete raw extraction

Unit	Form	Written	Printed	Raw	Conditional
Whitespace tokens	NFC	0.978970	0.924299	0.941117	1.643425
Alphanumeric runs	NFC	0.994126	0.850194	0.864808	1.656139
Non-space characters	NFC	0.887834	0.760641	0.776772	1.471341
Collapsed characters	NFC	0.904151	0.787669	0.803984	1.501794
Collapsed UTF-8 bytes	NFC	0.872465	0.767703	0.782502	1.485940
Whitespace tokens	NFKC	0.982913	0.928022	0.944908	1.650044
Alphanumeric runs	NFKC	0.994126	0.850194	0.864808	1.656139
Non-space characters	NFKC	0.889181	0.761993	0.778127	1.475428
Collapsed characters	NFKC	0.905936	0.789377	0.805705	1.506315
Collapsed UTF-8 bytes	NFKC	0.921638	0.805353	0.821683	1.537991

Exact furniture removed

Unit	Form	Written	Printed	Raw	Conditional
Whitespace tokens	NFC	0.971549	0.917413	0.933907	1.631137
Alphanumeric runs	NFC	0.986085	0.843249	0.857575	1.642889
Non-space characters	NFC	0.877145	0.751470	0.767376	1.453649
Collapsed characters	NFC	0.894022	0.778833	0.794910	1.485012
Collapsed UTF-8 bytes	NFC	0.863096	0.759454	0.774044	1.470049
Whitespace tokens	NFKC	0.975501	0.921144	0.937706	1.637771
Alphanumeric runs	NFKC	0.986085	0.843249	0.857575	1.642889
Non-space characters	NFKC	0.878511	0.752838	0.768747	1.457749
Collapsed characters	NFKC	0.895824	0.780554	0.796645	1.489544
Collapsed UTF-8 bytes	NFKC	0.911761	0.796712	0.812812	1.521556

Furniture removed and eight bracket events calibrated

Unit	Form	Written	Printed	Raw	Conditional
Whitespace tokens	NFC	0.971549	0.917413	0.933907	1.631137
Alphanumeric runs	NFC	0.986085	0.843249	0.857575	1.642889
Non-space characters	NFC	0.877145	0.751470	0.767376	1.453649
Collapsed characters	NFC	0.894022	0.778833	0.794910	1.485012
Collapsed UTF-8 bytes	NFC	0.863096	0.759454	0.774044	1.470049
Whitespace tokens	NFKC	0.975501	0.921144	0.937706	1.637771
Alphanumeric runs	NFKC	0.986085	0.843249	0.857575	1.642889
Non-space characters	NFKC	0.878511	0.752838	0.768747	1.457749
Collapsed characters	NFKC	0.895824	0.780554	0.796645	1.489544
Collapsed UTF-8 bytes	NFKC	0.911761	0.796712	0.812812	1.521556

Exact furniture removal deletes only isolated page numbers and repeated running headers. It keeps the source’s arXiv banner. The calibrated sensitivity replaces eight font- and occurrence-bound square-bracket components by their visible Unicode counterparts, retaining one component per event. Here that changes text hashes but none of the five metric counts. Printed hyphens remain literal.